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# -----  
# Cell 1 : Import Libraries  
# -----  
  
import pandas as pd  
import numpy as np  
import matplotlib.pyplot as plt  
  
from sklearn.model_selection import train_test_split  
from sklearn.linear_model import LinearRegression  
from sklearn.metrics import mean_squared_error, r2_score
```

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# -----  
# Cell 2 : Load Dataset  
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```
df = pd.read_csv("BostonHousing.csv")  
  
print("Dataset Loaded Successfully\n")  
  
print(df.head())
```

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# -----  
# Cell 3 : Dataset Information  
# -----
```

```
print("\n===== DATASET INFO =====\n")  
  
print(df.info())  
  
print("\n===== STATISTICAL SUMMARY =====\n")  
  
print(df.describe())
```

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# -----  
# Cell 4 : Check Missing Values  
# -----
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```
print("\n===== MISSING VALUES =====\n")  
  
print(df.isnull().sum())
```

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# -----  
# Cell 5 : Define Features and Target Variable  
# -----
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```
X = df.drop('medv', axis=1) # Features  
y = df['medv']             # Target Variable
```

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# -----  
# Cell 6 : Split Dataset into Training and Testing  
# -----
```

```
X_train, X_test, y_train, y_test = train_test_split(  
    X, y, test_size=0.2, random_state=42  
)
```

```
print("\nTraining Data Shape:", X_train.shape)  
print("Testing Data Shape:", X_test.shape)
```

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# -----  
# Cell 7 : Train Linear Regression Model  
# -----
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```
model = LinearRegression()
```

```
model.fit(X_train, y_train)
```

```
print("\nModel Trained Successfully")
```

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# -----  
# Cell 8 : Predict Test Data  
# -----
```

```
y_pred = model.predict(X_test)
```

```
print("\n===== PREDICTED VALUES =====\n")
```

```
print(y_pred[:10])
```

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# -----  
# Cell 9 : Model Evaluation  
# -----  
  
mse = mean_squared_error(y_test, y_pred)  
  
r2 = r2_score(y_test, y_pred)  
  
print("\n===== MODEL EVALUATION =====\n")  
  
print("Mean Squared Error :", mse)  
  
print("R2 Score :", r2)
```

```
# -----  
# Cell 10 : Actual vs Predicted Graph  
# -----  
  
plt.scatter(y_test, y_pred)  
  
plt.xlabel("Actual Prices")  
  
plt.ylabel("Predicted Prices")  
  
plt.title("Actual vs Predicted Prices")  
  
plt.show()
```